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The Editor
BMC Bioinformatics

Dear Dr Bendale,

I am pleased to submit the revised manuscript "Residual-Stream Geometry of Single-Cell Foundation Models Carries Incremental Gene-Regulatory Signal Across Tissues" (submission 2222a497-3fe7-4af2-bb39-d283ab0df022) for further consideration as a Research Article in *BMC Bioinformatics*.

I thank the editor and both reviewers for a constructive and detailed assessment. I have carried out a major revision that addresses every comment with **new experiments** rather than text changes alone, and have then recalibrated the manuscript's claims in light of the new results. A separate point-by-point response accompanies this submission.

The principal changes are as follows:

- **Leakage-resistant evaluation.** New leave-TF-out, leave-target-out and leave-both-out grouped cross-validation. The signal is robust to the first two but collapses under leave-both-out; this honest ceiling is now reported in the Results, Discussion and Limitations.
- **Comparable cross-model extraction.** Geneformer per-layer residual-stream activations were extracted from real forward passes, so the identical geometric pipeline is applied to both models; the previously apparent architectural gap is shown to be largely a representation-construction artifact.
- **Absolute metrics and multiplicity.** Absolute AUROC and AUPRC are now reported in every results table; the full grid of comparisons is Benjamini–Hochberg corrected; uncertainty is decomposed into within-run and across-seed components.
- **Robustness and confirmatory testing.** Harder degree- and expression-matched negative edges; a frozen, pre-specified pipeline applied once to held-out confirmatory domains.
- **New analyses requested by the reviewers.** Directionality from asymmetric geometric features; ensembling with expression-based GRN inference (GENIE3, co-expression); top-k retrieval/utility evaluation.
- **Recalibrated claims.** The title, abstract, contributions, Discussion and Conclusions now describe the result as incremental signal for retrospective edge prioritization; the Limitations were expanded from five to seven points; the cell-type interpretation is explicitly framed as a hypothesis.

All new analysis code and reproduction commands have been added to the public repository. The manuscript has not been published or submitted elsewhere, and the author has no competing interests to declare.

Sincerely,

Ihor Kendiukhov
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